Test Plan & Strategy

TP-001: Strategic Test Plan - Hilton Honors Quality Audit

1. Objective & Mission

The primary objective of this audit is to validate the **E2E User Conversion Funnel** for the Hilton Honors platform. The strategy prioritizes **Internationalization (i18n)** resilience and **Session Integrity** to ensure the "Golden Path"—from registration to checkout—is robust across diverse user locales and secure against PII leakage.

2. Quality Scope

Domain	In-Scope Activities
Functional	Full User Lifecycle, UTF-8 Encoding compliance, Search Logic (BVA), and API-to-UI Data Persistence.
Technical/Security	Cache Invalidation protocols, Session Token expiration, and XHR/Fetch payload inspection via Chrome DevTools.
Compatibility	Cross-engine validation: Chromium (Chrome) vs. WebKit (Safari) on macOS Sonoma.
Out of Scope	PCI-DSS compliant payment processing, Mobile Native applications, and Load/Stress performance testing.

3. Methodology & Test Techniques

This audit employs a **Risk-Based Testing (RBT)** approach using the following techniques:

- **Boundary Value Analysis (BVA):** Specifically applied to date-picker chronology and guest occupancy limits.
- **Negative Testing:** Targeted at input sanitization for password complexity and non-standard character sets.
- **Exploratory Testing:** Focused on "state-transition" anomalies (e.g., navigating back/forward during active sessions).

- **Technical Auditing:** Monitoring the Network layer to distinguish between client-side validation gates and server-side responses.

4. Environment & Data Management

- **Platform:** Production-equivalent Web Environment.
- **Browsers:** Chrome v122 (Primary), Safari v17.2 (Secondary).
- **Test Data Persona Alpha:** Standard ASCII user for baseline validation.
- **Test Data Persona Beta:** UTF-8 / Multi-byte user (e.g., `Díaz Ola`) for i18n stress-testing.

5. Quality Gates (Entry & Exit Criteria)

- **Entry Criteria:**
  1. Public-facing enrollment and search endpoints are reachable.
  2. Test credentials and unique email aliases are provisioned.
- **Exit Criteria:**
  1. 100% of "Critical" and "High" priority test cases executed.
  2. No "P0" (Blocker) defects remain open without a documented mitigation plan.
  3. Traceability matrix completed for all 11 test cases.

6. Risk Assessment & Mitigation

Risk	Impact	Mitigation Strategy
Limited Backend Access	High	Use **Black-Box API Inference** via Network Tab analysis to validate data payloads.
Dynamic Inventory Shifts	Low	Perform search validation across multiple properties to confirm consistent behavior.
PII Exposure	Critical	Enforce strict session cleanup and private browsing modes for all execution cycles.

7. Deliverables

- **Audit Suite:** 11 Traceable Test Cases.
- **Defect Reports:** Technical breakdown of the UTF-8 encoding failure (BUG-001).
- **Executive Summary:** Data-driven assessment of release readiness and international market risks.